

What is the DOM?

Introduction

The **DOM (Document Object Model)** is a programming interface provided by the browser that represents an HTML or XML document as a structured tree of objects. Each element, attribute, and piece of text in the HTML document becomes a **node** in the DOM tree.

This allows JavaScript to interact with the HTML and CSS of a web page — you can use JavaScript to read and modify the page's structure, content, and style dynamically.

Key Points:

- The DOM is **not** part of JavaScript, but it is provided by the browser's **Web APIs**.
- The browser turns HTML into a tree structure called the DOM.
- JavaScript can use this tree to:
 - Select elements
 - Modify content or style
 - Add or remove elements
 - Respond to user interactions

Example HTML

```
<!DOCTYPE html>
<html>
  <head>
    <title>DOM Example</title>
  </head>
  <body>
```

```
<h1>Hello, DOM!</h1>
<p>This is a paragraph.</p>
</body>
</html>
```

How the DOM Sees It

The browser parses the above HTML and creates a tree-like structure:

- document
 - html
 - head
 - title
 - body
 - h1
 - p

Accessing the DOM in JavaScript

The `document` object is the entry point to the DOM in JavaScript.

```
console.log(document); // Logs the entire DOM tree
console.log(document.body); // Logs the <body> element
console.log(document.title); // Logs the content of <title>
```

Summary

- The DOM represents the page so that JavaScript can interact with it.
- You can access and modify HTML elements using JavaScript through the DOM.
- Understanding the DOM is essential for web development and dynamic page interactions.